

財政學 HW1

1. (Hyman 10.1) Suppose you currently earn taxable income of \$100,000 per year. You are subject to an MTR of 50 percent. Currently, your ATR is 35 percent. Calculate your annual tax. Calculate the extra tax that you would pay per year if your annual income increased to \$110,000. What is your ATR when your annual income is \$110,000?

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Annual tax is \$35,000. The extra tax on \$10,000 additional income is \$5,000. The average tax rate on \$110,000 would be $\$40,000/\$110,000 = 36.36$ percent.

2. (Hyman 10.2) The payroll tax for unemployment insurance in a certain nation taxes all wages up to a maximum of \$30,000 per worker at a 5 percent flat rate. What are the marginal and average tax rates on the wages for each of the following three workers?

- a. A restaurant worker with annual wages of \$18,000.
- b. An assistant bank manager with wages of \$35,000 per year.
- c. A corporate CEO with an annual salary of \$500,000.

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- a. The restaurant worker has an average and marginal tax rate of 5 percent.
- b. The average tax rate for the bank manager is $\$1,500/\$35,000 = 4.29$ percent. The manager's marginal tax rate is zero.
- c. The average tax rate for the CEO is $\$1,500/\$500,000 = 0.3$ percent. The marginal tax rate for the CEO is zero.

3. (Hyman 10.3) A large city currently provides free water service to residents. The marginal social cost of making a gallon of water available per month is estimated to be 5 cents no matter how much water is used. Currently, city residents consume 500,000 gallons of water per month. The costs of making the water available are financed by a local tax on city residents.

- a. Draw a graph to show that the current monthly consumption of water is not efficient.
- b. Show the net gains in well-being possible by applying a user charge of 5 cents per gallon to residential users. Assume that monthly consumption declines to 400,000

gallons after the user charge is imposed. Calculate the tax revenues that can be freed for other uses each month (including a reduction in taxes to local residents) after the user charge is imposed.

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a. Because the price is zero, water will be consumed up to the point at which its marginal benefit is zero. At that level of output, the marginal cost will be five cents, so more than the efficient amount of water will be consumed because the marginal cost of water exceeds its marginal benefit.

b. The user charge of five cents will reduce quantity demanded until marginal benefit equals five cents. At that level of output, net gains from water production will increase by $0.5(100,000) \times 5 \text{ cents} = \$2,500$ per month, which is equal to the welfare triangle. Tax revenues of \$5,000 per month will be freed, because water production will fall by 100,000 gallons per month. In addition, the user charge will collect \$20,000 monthly from the sale of 400,000 gallons of water each month. A total of \$25,000 in tax revenue will therefore be freed as a result of the user charge.

4. (Rosen 14.8) Assume that in a given country, tax revenues, T , depend on income, I , according to the formula $T = -4000 + 0.2I$. Thus, for example, when a household has an income of \$50,000, its tax burden is $-4,000 + 0.2 \times 50,000$, or \$6,000.

a. Is this a progressive tax schedule? [Hint: Compute average tax rates at several different levels of income.]

b. Now let's generalize the tax schedule in this problem to: $T = a + tI$ where a and t are numbers. (For example, in the tax schedule above, $a = -4,000$ and $t = 0.2$.) Write down a formula for the average tax rate as a function of the level of income. Show that the tax system is progressive if a is negative, and regressive if a is positive. [Hint: The average tax rate is T/I .]

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The average tax rate equals $ATR = (-4000/I) + 0.2$ for any income level. Clearly this average tax rate converges to $ATR = 20\%$ as income gets large, and is lower for lower income levels. The tax system is progressive.

<i>Income</i>	<i>Tax Liability</i>	<i>Average Tax Rate</i>	<i>Marginal Tax Rate</i>
\$2,000	\$-3,600	-1.80	0.2
3,000	\$-3,400	-1.13	0.2
5,000	\$-3,000	-0.60	0.2
10,000	\$-2,000	-0.20	0.2
30,000	\$2,000	0.066	0.2

To show that the tax system is progressive if a is negative, write ATR as:

$$\text{ATR} = T/I = (a + TI)/I = a/I + t$$

Then take the derivative of ATR with respect to I in order to check whether ATR goes up or down as I increases: $d(\text{ATR})/dI = d(a/I + t)/dI = -a/I^2$. The change in average tax rate is increasing in income (which implies a progressive tax schedule) as long as a is negative.

5. (Rosen 14.13) Consider a society with only two people, one with an income of \$200,000 and one with an income of \$20,000. Assume that under the current tax system, the rich person pays \$50,000 in taxes and the poor person pays \$1,000. Suppose Congress passes a law that cuts the rich person's taxes by \$2,000 and the poor person's taxes by \$200. Using each of the equations below (Equ. A and Equ. B), assess whether this tax change increases or decreases progressivity.

Equ. A: $v_1 = \frac{\frac{T_1}{I_1} - \frac{T_0}{I_0}}{\frac{I_1}{I_0} - 1}$	Equ. B: $v_2 = \frac{\frac{T_1 - T_0}{T_0}}{\frac{I_1 - I_0}{I_0}}$
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(Suppose T_0 and T_1 are the true tax liabilities at income levels I_0 and I_1 , and $I_1 > I_0$.)

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Using Equ. A, the progressivity of the old tax system is:

$$0.00111 [= (50/200 - 1/20)/(200-20)],$$

and the progressivity of the new tax system is exactly the same

$$[= (48/200 - .8/20)/(200-20)].$$

Using Equ. B,

the progressivity of the old tax system is: 5.44 $[= ((50-1)/1)/((200-20)/20)],$

and the progressivity of the new tax system is: 6. $[= ((48-.8)/.8)/((200-20)/20)].$

Thus, under this measure of progressivity, the new tax system is more progressive.

6. MULTIPLE CHOICE QUESTIONS

(1) The tax base of a payroll tax is:

- a. consumer expenditures.
- b. interest income.
- c. labor income.

- d. both (b) and (c) are correct.

(2) An excise tax is a:

- a. general consumption tax.
- b. selective consumption tax.
- c. general wealth tax.
- d. selective tax on wealth.

(3) A proportional income tax has an average tax rate that:

- a. is always less than the marginal tax rate.
- b. always exceeds the marginal tax rate.
- c. equals the marginal tax rate at first and then becomes less than the marginal tax rate.
- d. always equals the marginal tax rate.

(4) A payroll tax taxes a worker's wages at 14 percent until the worker earns \$60,000 per year. All labor earnings in excess of \$60,000 are not subject to tax. The tax rate structure of the payroll tax is therefore:

- a. proportional.
- b. progressive.
- c. regressive.
- d. flat-rate.

(5) A government prints money to finance its expenditures. As a result,

- a. the economy can operate at a point outside its production possibility curve.
- b. inflation will occur.
- c. consumers will give up private goods to finance the increased government expenditures.
- d. both (b) and (c) are correct.

(6) Taxes:

- a. are voluntary payments to governments.
- b. are unlikely to affect market supply and demand.
- c. never affect efficiency in the allocation of resources.
- d. are compulsory payments associated with certain activities.

(7) The marginal tax rate will eventually exceed the average tax rate for a:

- a. proportional tax.

- b. regressive tax.
- c. progressive tax.
- d. flat-rate tax.

(8) If the marginal tax rate is 20 percent under a proportional tax rate structure, the average tax rate:

- a. should be 20 percent.
- b. should be above 20 percent.
- c. should be below 20 percent.
- d. cannot be determined.

(9) If the average tax rate under a progressive tax rate structure is 35 percent, a possible marginal tax rate is:

- a. 30 percent.
- b. 25 percent.
- c. 42 percent.
- d. cannot be determined.

(10) A 5-percent retail sales tax on all consumer purchases in a state is imposed. The sales tax is:

- a. a flat-rate tax.
- b. a tax with a regressive rate structure.
- c. levied on an income base.
- d. all of these are correct.

7. 舉出三項台灣目前的指定用途稅，其用途各自為何？

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參見立法院全球資訊網

https://www.ly.gov.tw/Pages/ashx/File.ashx?FilePath=~/File/Attach/186408/File_230316.doc

8. 請查詢台灣110年度綜合所得稅平均稅率及有效稅率(五分位數)

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參見110年度綜合所得稅申報核定統計專冊

<https://www.fia.gov.tw/WEB/fia/ias/isa110/ISA110.html> (表6 綜合所得、應納稅額及稅率統計表 3-0)

表6-3-0(110年度)綜合所得應納稅額及稅率5分位申報統計表

單位：金額(千元)
版號：01112024

分位別	納稅單位	綜合所得總額	所得淨額		應納稅額		稅後所得		平均稅率	有效稅率
			金額	百分比	金額	百分比	金額	百分比		
第1分位	1289738	108063078	1062	0.00	53	0.00	108063025	100.00	5.00	0.00
第2分位	1289738	349308301	33320053	9.54	1665793	0.48	347642508	99.52	5.00	0.48
第3分位	1289738	611853028	124187374	20.30	6209042	1.01	605643985	98.99	5.00	1.01
第4分位	1289738	1051275918	322904211	30.72	17445104	1.66	1033830814	98.34	5.40	1.66
第5分位	1289737	2976639038	1900469782	63.85	263890440	8.87	2712748599	91.13	13.89	8.87
合計	6448689	5097139363	2380882482	46.71	289210433	5.67	4807928930	94.33	12.15	5.67